Ans. (b) : The surface area of a cube $= 6a^2 = 216$

$$\Rightarrow a^2 = \frac{216}{6} = 36$$

$$\Rightarrow \boxed{a = 6}$$

Volume of a cube $= a^3$

$= 6^3$

$= 216\ m^3$

138. If Rs. 1000 is invested at 12% interest and interest is compounded half yearly, what will be the total amount at the end of one year ?

(a) Rs. 1120.00 (b) Rs. 1123.60

(c) Rs. 1126.20 (d) Rs. 1134.40

Ans. (b) :

Principal (p) = 1000 Rs.

Rate (r) = 12% yearly or 6% half yearly

Time (t) = 1 year

So, amount (A) $= 1000\left(1+\frac{6}{100}\right)^2$

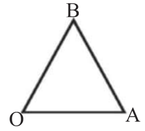
$$= 1000 \times \frac{53\times53}{50\times50} = 1123.60\,\text{Rs.}$$

139. Two persons start walking at a steady pace of 3 km/hour from a road 1 intersection along two roads that make an angle of 60° with each other. What will be the (shortest) distance separating them at the end of 20 minutes ?

(a) 3 km (b) 2 km

(c) 1.5 km (d) 1 km

Ans. (d) :



After 20 minute the distance $= \frac{1}{3}\times3 = 1$

From ΔAOB,

$\angle AOB = 60^0$

So, AOB is a Isosceles triangle in which distance is 1 km.

140. Which one of the following figures will generate a cone when rotated about one of its straight edges ?

(a) An equilateral triangle

(b) A sector of a circle

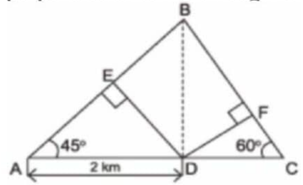
(c) A segment of a circle

(d) A rightangled triangle

Ans. (d) : When a right angled triangle is rotated on one of its straight sides, the shape of a cone will be formed. This can be clearly understood by the following picture.



141. A person at a point D on a straight road AC has four options to go to B which lies on a perpendicular to AC through D.



Which one of the following is the shortest route to B ?

(a) D to A and A to B

(b) D to E and E to B

(c) D to C and C to B

(d) D to F and F to B

Ans. (d) : From the figure, in the right angled triangle DAE

$$\sin 45^0 = \frac{DE}{AD}$$

So, $DE = AD \times \sin 45^0$

Or $DE = 2\times\frac{1}{\sqrt{2}}$

$$= \frac{\sqrt{2}\times\sqrt{2}}{\sqrt{2}}$$

$$= \sqrt{2}$$

Same as in ΔDBF-

$$\sin 30^0 = \frac{DF}{BD}$$

So, $DF = BD \times \sin 30^0 = BD \times \frac{1}{2} = \frac{2}{2} = 1$ km

So, if a person move from point D to E and point F to B then road will be too short. So right answer is (d)

142. There are three tables containing two drawers each. It is known that one of the table contains a silver coin in each of its drawers, another table contains a gold coin in each drawer while the third table contains a silver coin in one drawer and a gold coin in the other. One of the drawers of a table is opened and found to contain a silver coin. What is the probability that the other drawer of that table contains a gold coin ?

(a) 1 (b) 0.5

(c) 0.25 (d) 0.75

Ans. (b) : There are three tables containing two drawers each

i.e. $\boxed{S\ S}$ $\boxed{G\ G}$ $\boxed{S\ G}$

here S = silver coin

G = Gold coin

We know that one of the table contains a silver coin in each of its drawers, another table contains a gold coin in each drawer while the third table contains a silver coin in one drawer and a gold coin in the other. One of the drawers of a table is opened and found to contain a silver coin.

So, the probability of other drawer to contain gold coin $= \frac{1}{2} = 0.5$

143. In the sequence 462, 420, 380, X, 306. X stands for

(a) 352 (b) 342

(c) 332 (d) 322

Ans. (b) : There is a difference of 2 respectively in the consecutive terms,

$\because 462 - 420 = 42$

$420 - 380 = 40$

$380 - 342 = 38$

$342 - 306 = 36$

$\therefore x = 380 - 38 = 342$ Ans.

144. Ram and Shyam work on a job together for four days and complete 60% of it. Ram takes leave then and Shyam works for eight more days to complete the job. How long would Ram take to complete the entire job alone?

(a) 6 days (b) 8 days

(c) 10 days (d) 11 days

Ans. (c) : Ram and Shyam work in 4 days = 60% or 3/5 (parts)

$\therefore$ Ram and Shyam's work in 1 day $= \frac{3}{20}$

$\therefore$ Remaining work after 4 days $= 1-\frac{3}{5} = \frac{2}{5}$

$\therefore$ Shyam work in 8 days $= \frac{2}{5}$

$\therefore$ Shyam done total work $= 8\times\frac{5}{2} = 20$ days

$\therefore$ Shyam's work in 1 days $= \frac{1}{20}$ part

Hence, Ram's work in 1 day $= \frac{3}{20}-\frac{1}{20} = \frac{1}{10}$

Hence, Ram will do that work = 10 days

145. The square root of the cube of this number is the cube of its square root. It is not 1 and it is less than 6. What is it ?

(a) 2 (b) 3

(c) 4 (d) 5

Ans. (c) : By option, given number = 4 (let)

then, $\sqrt{(4^3)} = \left(\sqrt{4}\right)^3$

$\Rightarrow \sqrt{64} = (2)^3$

$\Rightarrow \boxed{8 = 8}$ proved hence the number will be 4.

146. Ten per cent of twenty plus twenty per cent of ten equals

(a) 10 per cent of 20 (b) 20 per cent of 200

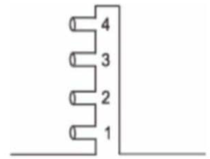
(c) 1 per cent of 200 (d) 2 per cent of 200

Ans. (d) : From the question,

$$20\times\frac{10}{100}+10\times\frac{20}{100} = 2+2 = 4$$

$= 2\%$ of 200

147. Consider the figure of a fountain with four holes given below :



Which one of the holes in the fountain will throw the water farthest ?

(a) 4 (b) 3

(c) 2 (d) 1

Ans. (c) : According to the application of Bernoulli's theorem, if a hole is made in the middle of a tank full of water, then the stream of water will fall the farthest. But there is no hole in the middle of the fountain, so the pressure of the water column at hole 2 will be higher than at hole 3, hence stream will be thrown up to the maximum distance from the hole 2.

148. If $\boxed{B}\ \boxed{J}\ \boxed{P}$ is $\boxed{D}\ \boxed{M}\ \boxed{K}$ then $\boxed{R}\ \boxed{S}\ \boxed{S}$ would be

(a) $\boxed{C}\ \boxed{P}\ \boxed{I}$

(b) $\boxed{T}\ \boxed{V}\ \boxed{N}$

(c) $\boxed{S}\ \boxed{J}\ \boxed{P}$

(d) $\boxed{T}\ \boxed{D}\ \boxed{P}$

Ans. (b) : From question-

Given, $B \xrightarrow{+2} D$ So, $R \xrightarrow{+2} T$

$J \xrightarrow{+3} M$ $S \xrightarrow{+3} V$

$P \xrightarrow{-5} K$ $S \xrightarrow{-5} N$

Ans- TVN

149. The letters L, M, N, O, P, Q, R, S and T in their order are substituted by nine integers 1 to 9 but not in that order, 4 is assigned to P. The difference between P and T is 5. The difference between N and T is 3. What is the integer assigned to N?

(a) 7 (b) 5

(c) 4 (d) 6

Ans. (d) : From question

L	M	N	O	P	Q	R	S	T
	6		4					9

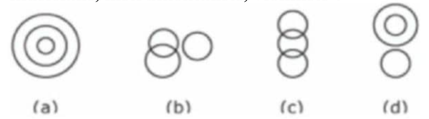
$\because$ Difference between P & T = 5

$\therefore T = P + 5 = 4 + 5 = 9$

$\because$ Difference between N & T = 3

$\therefore N = T - 3 = 9 - 3 = 6$ Ans.

150. Which one of the following four logical diagrams represents correctly the relationship between musicians, instrumentalists, violinists ?



Ans. (a) : Option (a) represents correctly the relationship between musicians, instrumentalists, violinists in logical diagrams.